Check for updates

ACM DIGITAL LIBRARY

Association for Computing Machinery

acm open

Latest updates: https://dl.acm.org/doi/10.1145/3638530.3654234

POSTER

Multi-objective multi-population Genetic Programming for feature selection and classification to high-dimensional data

QIAOMAN LI, Hebei Agricultural University, Baoding, Hebei, China

XIAOYING GAO, Victoria University of Wellington, Wellington, WGN, New Zealand

WENYANG MENG, Hebei Agricultural University, Baoding, Hebei, China

JIANBIN MA, Hebei Agricultural University, Baoding, Hebei, China

Open Access Support provided by:

Hebei Agricultural University

Victoria University of Wellington

PDF Download
3638530.3654234.pdf
26 February 2026
Total Citations: 3
Total Downloads: 361

Published: 01 August 2024

Citation in BibTeX format

GECCO '24 Companion: Genetic and Evolutionary Computation Conference Companion
July 14 - 18, 2024
VIC, Melbourne, Australia

Conference Sponsors:
SIGEVO

Multi-objective multi-population Genetic Programming for feature selection and classification to high-dimensional data

Qiaoman Li
1283798084@qq.com
College of Information Science and Technology, Hebei Agricultural University,
Baoding, China

Xiaoying Gao
xiaoying.gao@vuw.ac.nz
School of Engineering and Computer Science, Victoria University of Wellington,
Wellington, New Zealand

Wenyang Meng
1751797118@qq.com
College of Information Science and Technology, Hebei Agricultural University,
Baoding, China

Jianbin Ma*
majianbin@hebau.edu.cn
College of Information Science and Technology, Hebei Key Laboratory of Agricultural Big Data, Hebei Agricultural University,
Baoding, China

ABSTRACT

Classification for high-dimensional data is a challenging task due to a great number of redundant and irrelevant features. Genetic Programming (GP) has its built-in feature selection characteristics and is suitable for processing high-dimensional data. However, if the number of features in GP individual is not restricted, the bloat phenomenon will occur, which affects the generalization of the training model. Moreover, GP is easy to fall into local search in the evolutionary process. In this paper, a multi-objective multi-population GP classifier construction method is proposed, which uses a multi-population co-evolution strategy to divide the evolving population into the main and auxiliary populations with different evolutionary strategies and different evaluation criteria, and employs a multi-objective strategy to ensure classification performance and restrict the number of features at the same time. Experiments on seven high-dimensional datasets show that our proposed multi-population co-evolution strategy and multi-objective strategy are all effective to improve the classification performance of GP classifiers. Comparisons with three state-of-the-art GP classifier construction methods show that our proposed methods achieve better or comparable classification performance on most cases.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; *Bio-inspired approaches*; **Genetic programming**;

KEYWORDS

High-dimensional data, Multi-objective, Multi-population, Genetic programming

*Corresponding author.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
GECCO '24 Companion, July 14–18, 2024, Melbourne, VIC, Australia
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0495-6/24/07.
https://doi.org/10.1145/3638530.3654234

ACM Reference Format:
Qiaoman Li, Xiaoying Gao, Wenyang Meng, and Jianbin Ma. 2024. Multi-objective multi-population Genetic Programming for feature selection and classification to high-dimensional data. In *Genetic and Evolutionary Computation Conference (GECCO '24 Companion), July 14–18, 2024, Melbourne, VIC, Australia.* ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3638530.3654234

1 INTRODUCTION

Nowadays, there are a large number of high-dimensional applications. High-dimensional data involves cancer data, biological genetic data, image data, text data, etc. High-dimensional data contains a large number of redundant and irrelevant features, and these ineffective features bring the problem of "Curse of dimensionality" to classification. Feature selection is commonly used to deal with this problem. Genetic programming (GP) is an effective evolutionary computation algorithm, and can be used for feature construction and classifier construction [2] due to its flexible representation. Moreover, GP has its built-in feature selection characteristics and is suitable for processing high-dimensional data [10].

GP individuals consist of internal nodes and terminal nodes. For classification, the terminal nodes of GP individuals consist of original features. If the number of terminal nodes is not restricted, the evolved classification model will become increasingly large, which will result in the bloat phenomenon and affects the generalization of the GP classification model. Ma and Gao [2] proposed a classifier construction method that uses a multi-objective fitness function to reduce classification error rate and the number of selected features. Muni et al. [5] and Purohit et al. [7] used similar methods to restrict the number of selected features for feature selection and classification. However, it is difficult to balance the number of features and classification performance. Several studies have considered restricting the size of GP individuals while optimizing classification performance using multi-objective GP (MOGP). Nag and Pal [6] proposed an ensemble algorithm for simultaneous feature selection and classifier construction using MOGP. Shao et al. [8] developed an evolutionary learning method to automatically generate domain-adaptive global feature descriptors for image classification using MOGP. Few studies have investigated whether using MOGP can

reduce the number of features while improving classification performance. Therefore, this paper adopts the idea of multi-objective evolution strategy to restrict the number of selected features while improving the classification performance of the GP classifier and investigates the effectiveness through experiments.

Evolutionary algorithms tend to fall into local search, resulting in premature convergence and local search. Multi-population co-evolution can solve the problem of falling into local search to some extent. Multi-population co-evolution can maintain multi-population simultaneously, and each of which can explore different regions of the search space. This search strategy increases the diversity of the search, helps to find the global optimal solution, and avoids falling into the local search. Shi et al. [9] proposed a strategy for predicting RNA secondary structure, which involves multi-population to evolve together in a cooperative manner in each iteration. Ma and Bai [4] proposed an improved differential evolution of multi-population with an optimal stochastic mutation strategy. The population is divided into three sub-populations based on fitness values, and each sub-population uses different mutation strategies and control parameters, and individuals share different mutation strategies and control parameters by migrating between sub-populations. Zhu et al. [13] proposed an adaptive multi-population differential evolution algorithm. The population was randomly divided into three equal-sized sub-populations, each with different mutation strategies. At the end of each generation, all subpopulations are updated and reorganized independently. Zhao et al. [12] proposed an improved NSGA-II algorithm, which divides the original population into multi-population and assigns a different crossover operator to each subspecies. Multi-population co-evolution strategy has been used in differential evolution algorithm, Genetic Algorithm, etc., but there are still few studies about multi-population co-evolution on GP classifier construction. Whether multi-population co-evolution can further improve the classification performance should be investigated further.

In this paper, to deal with high-dimensional data, a GP classifier construction method based on multi-objective and multi-population strategy for simultaneous feature selection and classification is proposed to improve GP's generalization ability and local search. The method combines GP and the multi-objective evolutionary strategy of NSGA-II as well as the multi-population co-evolution strategy. The multi-objective strategy can simultaneously improve the performance of the classifier as well as reduce the number of selected features. The multi-population co-evolutionary strategy avoids GP falling into local search. The effectiveness of the proposed method is verified through experiments.

2 METHODOLOGY

2.1 Fitness Functions

Bhowan et al. [1] investigated fitness functions including AUC approximation (Wmw) and Average Mean Square Error (AMSE) for solving unbalanced classification problems and experiments show that Wmw and AMSE can achieve good classification performance. Our proposed method employs Wmw and AMSE to evalate the classificaton performance of GP classifiers and use the number of features to restrict the number of selected features. The larger the AMSE value and the AUC approximating (Wmw) value, the better the classification performance of a GP classifier. The smaller the number of features, the simpler a GP classifier.

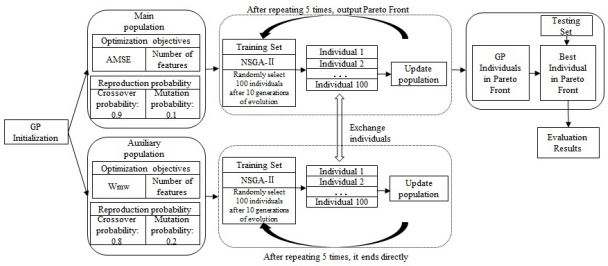

Figure 1: The framework of multi-objective multi-population GP

2.2 Method Overview

Our multi-population co-evolution strategy is used to divide the evolving population into the main and auxiliary polulations. The main purpose of regular migration of individuals between the two populations is to inject new individuals into the population in order to prevent too much duplication of individuals in evolution. In the process of co-evolution of the main and auxiliary populations, different crossover and mutation probabilities as well as different fitness functions are assigned to different populations in order to prevent too much overlap individuals.

The framework of multi-objective multi-population GP is shown in Figure 1. In the process of evolution, the setting of crossover and mutation probabilities is crucial as they directly affect the behavior and performance of the algorithm. According to the relevant parameter settings from literatures [3] and [6], in the main population, the crossover probability is 90% and the mutation probability is 10%; in the auxiliary population, the crossover probability is 80% and the mutation probability is 20%. By setting different crossover and mutation probabilities for the main and auxiliary populations, we can exchange, recombine, or mutate individuals in different ways between the two populations, thereby increasing population diversity. Average mean square error (AMSE) and the number of features are used as the fitness functions for the main population, and approximate AUC (Wmw) and the number of features are used as the fitness functions for the auxiliary population. Different fitness functions can control the evolutionary direction of individuals and avoid a large number of duplications of individuals in different populations. A large number of experiments prove that setting the generations of migrating to 10, and the number of migrating individuals to 100 can achieve good classification performance. Therefore, 100 migrating individuals are randomly selected from both the main and auxiliary populations after 10 generations of evolution and exchanged, and the individuals in the main and auxiliary populations are updated after the migration is completed. The updated main and auxiliary populations are re-entered into the evolutionary process. After evolution is complete, the individual with the highest AMSE in the pareto front of the main population is selected as the optimal individual.

3 EXPERIMENTS

3.1 Datasets and Experimental Design

Seven high-dimensional cancer gene datasets were collected from the website (http://csse.szu.edu.cn/staff/zhuzx/software.html) to verify the effectiveness of our proposed method. The dataset includes Colon, Dlbcl, Leukemia, Lymphoma, MLL, ovarian, and Srbct. All datasets are high-dimensional small-sample datasets with thousands or tens of thousands of features. These datasets include four binary-class datasets and three multi-class datasets. Multi-class datasets are transformed to binary-class dataset, where one category in the multi-class dataset is merged with other categories to create a binary-class dataset for experiments.

Since evolutionary computation has a stochastic nature, in order to avoid the impact of stochastic characteristics of GP and dataset partitions on the experimental results, the average of 30 experimental results is taken as the final experimental results. Each experiment randomly divides the dataset into a training set and a testing set, of which, 70% is the training set and 30% is the testing set. The training set is used to train the classification model and the testing set is used to test the classification effect of the classification model. The AUC (%) is used to evaluate the classification performance.

The function set of GP individuals consists of $+$, $-$, $\times$, % (protective division). The terminal set consists of the original features in the dataset. The ramped half-and-half is used to initialize individuals. The maximum depth of the individual (tree) is 8. Individuals are selected using the tournament selection method. The population size is set to 500. The number of generations is set to 50.

3.2 Experimental Results and Analysis

In order to verify whether both multi-objective and multi-population approaches can improve the experimental results, the effectiveness of the proposed approach is verified by the following classifier variants, including the classifier based on single population and single objective (SSGP), the classifier based on multi-population and single objective (MSGP) and the proposed multi-objective multi-population classifier (MMGP). The SSGP uses AMSE as the evaluation criterion. The MSGP uses AMSE as the evaluation criterion of the main population and Wmw as the evaluation criterion of the auxiliary population, and the MMGP uses AMSE and the number of features as the evaluation criterion of the main population and Wmw and the number of features as the evaluation criterion of the auxiliary population. The SSGP and the MSGP are compared in order to verify whether multi-population can improve the experimental results, and MSGP and MMGP are compared to verify whether multiple objectives can improve the experimental results.

3.2.1 Comparisons of MMGP, MSGP, and SSGP. The experimental results of SSGP, MSGP and MMGP are shown in Table 1. In the table, "Best" denotes the best AUC value of 30 independent runs, "A±Std" represents the average AUC and the corresponding standard deviation of 30 independent runs, and "#F" is the average number of features for 30 GP runs. As shown in Table 1, compared with SSGP, MSGP can improve the classification performance on 5 out of 7 datasets. MSGP achieves classification performance 4.78%, 6.34%, and 10.73% higher than SSGP on datasets Dlbcl, Lymphoma, and MLL, respectively. The experimental results verify that the multiple population co-evolution strategy can improve the population diversity of GP, which helps to find the global optimal solution and avoids falling into the local search. At the same time, compared to SSGP, MSGP can obtain a smaller number of features on all seven datasets, which shows that MSGP can improve the classification performance while limiting the size of the classification model, and therefore achieving the purpose of feature selection.

Table 1: The Experimental Results of SSGP, MSGP and MMGP

Dataset	Method	Best	A±Std	#F
Colon	SSGP	100.00	80.35±12.63	25.13
	MSGP	98.80	78.21±9.26	21.06
	MMGP	95.23	**80.95±8.69**	4.66
Dlbcl	SSGP	100.00	91.14±6.07	27.50
	MSGP	100.00	95.92±4.02	17.33
	MMGP	100.00	**99.47±1.61**	1.40
Leukemia	SSGP	99.10	93.33±16.45	21.90
	MSGP	100.00	93.66±7.50	12.00
	MMGP	100.00	**95.38±3.98**	2.66
Lymphoma	SSGP	100.00	85.09±9.06	20.40
	MSGP	100.00	91.43±10.98	5.46
	MMGP	100.00	**91.63±8.91**	1.90
MLL	SSGP	100.00	75.48±16.42	27.36
	MSGP	100.00	86.21±14.94	14.6
	MMGP	100.00	**88.99±10.58**	3.23
ovarian	SSGP	100.00	99.14±1.38	19.63
	MSGP	100.00	99.09±1.23	14.13
	MMGP	100.00	**99.69±0.50**	3.06
Srbct	SSGP	100.00	94.90±4.53	17.06
	MSGP	100.00	95.06±8.41	7.90
	MMGP	100.00	**98.99±1.65**	1.86

As shown in Table 1, compared to MSGP, MMGP can achieve better classification performance on all the seven datasets. The classification performance of MMGP is 2.74%, 3.55%, 1.72%, 2.78% and 3.93% higher than that of MSGP on datasets Colon, Dlbcl, Leukemia, MLL and Srbct. Moreover, MMGP considers the number of features as an optimization objective. From the experimental results in Table 1, we can see that MMGP only selects a few features, while achieving higher classification performance. The experimental results verify that our proposed MMGP can not only evolve effective classifiers but also achieve the purpose of feature selection.

3.2.2 Comparisons of MMGP and Three Benchmark Techniques. In order to verify the effectiveness of our proposed MMGP classifier construction method, three state-of-the-art GP classifier construction methods are selected for comparison. The three methods are Muni [5], Nag [6], and Wang [11]. The experimental results of MMGP and three benchmark techniques are shown in Table 2. "Wtest" denotes the Wilcoxon significant test with a significance level of 0.05. "+" or "−" indicates the experimental results of MMGP are significantly better or worse than other methods with p-value=0.05, "++" indicates the experimental results of MMGP are

Table 2: The Experimental Results of MMGP and Three Benchmark Techniques

Dataset	Method	A±Std	#F	Wtest
Colon	**MMGP**	**80.95±8.69**	4.66	
	Muni	72.61±0.14	16.63	=
	Nag	80.67±0.10	5.76	+
	Wang	71.86±0.13	19.16	++
Dlbcl	**MMGP**	**99.47±1.61**	1.4	
	Muni	87.80±0.14	13.6	++
	Nag	94.25±0.05	5.16	++
	Wang	88.36±0.10	15.46	++
Leukemia	**MMGP**	**95.38±3.98**	2.66	
	Muni	86.57±0.11	10.60	=
	Nag	94.73±0.04	4.26	++
	Wang	92.26±0.09	10.26	=
Lymphoma	**MMGP**	**91.63±8.91**	1.90	
	Muni	87.58±0.14	10.73	=
	Nag	90.26±0.10	4.43	=
	Wang	89.54±0.09	8.40	=
MLL	**MMGP**	88.99±10.58	3.23	
	Muni	67.50±0.16	15.06	=
	Nag	**91.31±0.09**	4.60	++
	Wang	79.34±0.14	16.30	++
ovarian	**MMGP**	**99.69±0.50**	3.06	
	Muni	95.43±0.06	18.83	++
	Nag	98.64±0.01	9.06	++
	Wang	99.14±0.01	15.63	+
Srbct	**MMGP**	**98.99±1.65**	1.86	
	Muni	94.73±0.07	8.03	+
	Nag	97.83±0.02	3.30	++
	Wang	95.66±0.06	5.00	++

significantly better than other methods with p-value=0.01, and "=" means the experimental results are similar.

As shown in Table 2, MMGP can achieve better classification performance than benchmark techniques on almost all the cases. According to the Wilcoxon significant test, MMGP can achieve significantly better classification performance than other benchmark techniques in 14 out of 3*7=21 cases and obtain similar classification performance in 7 out of 21 cases. Moreover, we can see that MMGP can obtain smaller number of selected features than other methods on all datasets. The comparisons with other benchmark techniques further confirm that our proposed method can achieve more competitive classification performance and obtain fewer features.

4 CONCLUSIONS

In this paper, a feature selection and classification method for high-dimensional data based on multi-objective multi-population GP is proposed, which employs two populations to co-evolve, and exchanges individuals during the evolutionary process to realize the information exchange between the populations in order to avoid local search. The main and auxiliary populations are evaluated using different evaluation criteria, and optimization objectives including Wmw and AMSE ensure classification performance, and the number of features as another optimization objective aims to select effective features. Experiments on seven high-dimensional cancer gene datasets show that our proposed method show good results. Compared to the GP classifier based on multi-population with single objective and the GP classifier based on single populations with single objective, the classification performance of the multi-objective multi-population GP classifier can achieve better classfication performance and obtain a smaller number of features, which indicates that our proposed method realizes the tasks of feature selection and classification at the same time. Comparisons with three state-of-the-art GP classifiers show that our proposed methods achieve better or comparable classification performance on most cases.

Classification for high dimensional data is a very important and challenging task. In this paper, we propose a multi-objective multi-population GP classifier construction method. Although we have achieved good experimental results, the following work can be carried out in the future. A dynamic migration strategy can be considered for population migration, i.e., adaptively adjusting the frequency and mode of migration according to the performance of the algorithm and the characteristics of the problem, and no longer exchanging individuals with a static crossover and mutation probability and a fixed frequency.

ACKNOWLEDGMENTS

This work was supported by the Hebei Key Laboratory of Agricultural Big Data, China.

REFERENCES

[1] Urvesh Bhowan, Mark Johnston, and Mengjie Zhang. 2011. Developing new fitness functions in genetic programming for classification with unbalanced data. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)* 42, 2 (2011), 406–421.

[2] Jianbin Ma and Xiaoying Gao. 2020. Designing genetic programming classifiers with feature selection and feature construction. *Applied Soft Computing* 97 (2020), 106826.

[3] Jianbin Ma and Guifa Teng. 2019. A hybrid multiple feature construction approach using Genetic Programming. *Applied Soft Computing* 80 (2019), 687–699.

[4] Yongjie Ma and Yulong Bai. 2020. A multi-population differential evolution with best-random mutation strategy for large-scale global optimization. *Applied Intelligence* 50, 5 (2020), 1510–1526.

[5] Durga Prasad Muni, Nikhil R Pal, and Jyotirmay Das. 2006. Genetic programming for simultaneous feature selection and classifier design. *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)* 36, 1 (2006), 106–117.

[6] Kaustuv Nag and Nikhil R Pal. 2015. A multiobjective genetic programming-based ensemble for simultaneous feature selection and classification. *IEEE transactions on cybernetics* 46, 2 (2015), 499–510.

[7] Anuradha Purohit, Narendra S Chaudhari, and Aruna Tiwari. 2010. Construction of classifier with feature selection based on genetic programming. In *IEEE Congress on Evolutionary Computation*. IEEE, 1–5.

[8] Ling Shao, Li Liu, and Xuelong Li. 2013. Feature learning for image classification via multiobjective genetic programming. *IEEE Transactions on Neural Networks and Learning Systems* 25, 7 (2013), 1359–1371.

[9] Sha Shi, Xin-Li Zhang, Xian-Li Zhao, Le Yang, Wei Du, and Yun-Jiang Wang. 2019. Prediction of the RNA secondary structure using a multi-population assisted quantum genetic algorithm. *Human Heredity* 84, 1 (2019), 1–8.

[10] Binh Tran, Bing Xue, and Mengjie Zhang. 2019. Genetic programming for multiple-feature construction on high-dimensional classification. *Pattern Recognition* 93 (2019), 404–417.

[11] Pu Wang, Ke Tang, Thomas Weise, Edward PK Tsang, and Xin Yao. 2014. Multiobjective genetic programming for maximizing ROC performance. *Neurocomputing* 125 (2014), 102–118.

[12] Zhibiao Zhao, Bin Liu, Chunran Zhang, and Haoran Liu. 2019. An improved adaptive NSGA-II with multi-population algorithm. *Applied Intelligence* 49 (2019), 569–580.

[13] Lin Zhu, Yongjie Ma, and Yulong Bai. 2020. A self-adaptive multi-population differential evolution algorithm. *Natural Computing* 19 (2020), 211–235.